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Hiring Manager

CGI

Dear Hiring Manager,

My name is Ajay Venkatesh. I'm finishing my M.S. in Computer Engineering at NYU, with production software experience at PwC in Bengaluru, a summer SDE internship at Amazon, and ongoing on-campus work at NYU's Novel AI Technologies. I'm writing about the Python Backend Engineer role at CGI.

**Python** is my daily driver for both backend services and ML pipelines. At Amazon I provisioned **Infrastructure-as-Code** on **AWS CDK** (Lambda, API Gateway, DynamoDB, IAM, CloudWatch) with multi-stage CI/CD and rollback controls, plus a high-throughput, event-driven processing system on **AWS ECS Fargate**, **SQS**, **SNS** that decoupled distributed message handling under variable load. At Campus Mesh I built scalable **Node.js microservices** powering real-time academic workflows backed by **MongoDB**, plus an LLM-powered AI assistant with **prompt engineering** and **RAG retrieval** over structured data.

On backend architecture: I'm comfortable designing REST APIs, working across relational and non-relational stores (**PostgreSQL**, **MySQL**, **MongoDB**, **DynamoDB**), and building production systems with monitoring, observability, and reliability under load. At PwC I owned a distributed backend on **Microsoft Azure** with event-driven processing, fault-tolerant retries, and parallelized batch ingestion that materially cut execution time on operational feeds.

On engineering discipline: I ship under code-review discipline with unit and integration testing as part of the build. At PwC I led code reviews and mentored engineers on software design principles, and at Amazon I used **Mockito** for backend testing of Java services. The same posture applies to Python development with **pytest** and standard linting workflows.

Stack-wise: **Python**, **Java**, **TypeScript**, **SQL**, plus **AWS**, **Docker**, **Kubernetes**, and **CI/CD** workflows. Comfortable across cloud platforms, distributed systems, and the testing discipline that consulting-style delivery demands.

What pulls me toward CGI is the operational seriousness of consulting-style work, where engineering across client engagements has to be both production-quality and shippable on real timelines, with documentation and accountability that compound across teams.

I'd welcome the chance to discuss further. Thanks for considering my application.

Sincerely,

Ajay Venkatesh